
MODEL SELECTION AND VALIDATION REPORT (remove_2_biomass model)

OBSERVATIONS:1724

SANITY: warning - sanity check failed for the following models:
spatiotemporal_random_fields_iid_with_anisotropy

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_2_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.68	0.30
remove_2_kde_fric	-0.06	0.02
remove_2_kde_feve	-0.29	0.03
sbt_mean	0.20	0.09
sbt_sd	-0.06	0.04
sst_sd	0.03	0.04
log_chla_mean	0.02	0.03
depth	0.14	0.04
log_sum_fishing	0.11	0.03

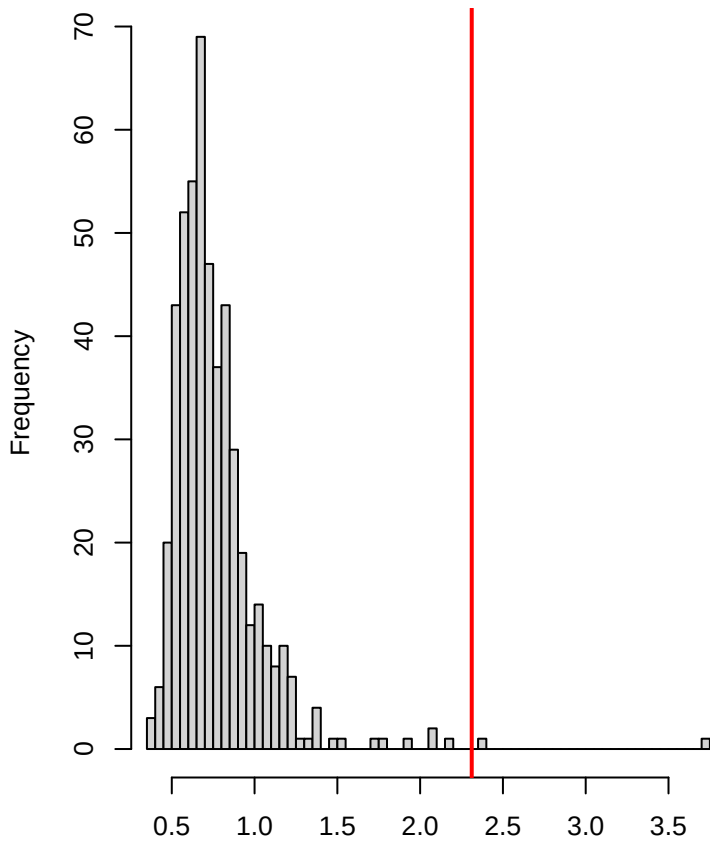
Dispersion parameter: 0.83
Spatiotemporal AR1 correlation (rho): 0.93
Matérn range: 361.37
Spatiotemporal marginal AR1 SD: 0.90
REML criterion at convergence: 13053.162

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

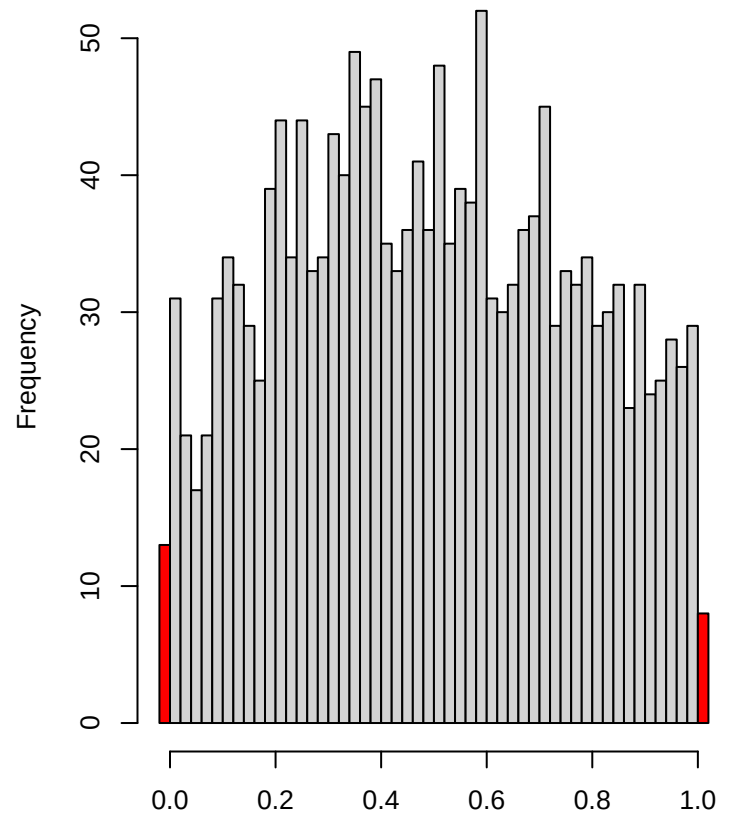
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	26132.32	2.5627697
spatiotemporal_random_fields_ar1_with_anisotropy	15	26134.89	144.6416672
spatiotemporal_random_fields_iid	12	26279.53	0.3327776
spatial_random_fields	12	26279.86	0.1569295
spatial_random_fields_with_anisotropy	14	26280.02	118.3960056
spatiotemporal_random_fields_iid_with_anisotropy	14	26398.41	131.6792779
error_only	10	26530.09	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



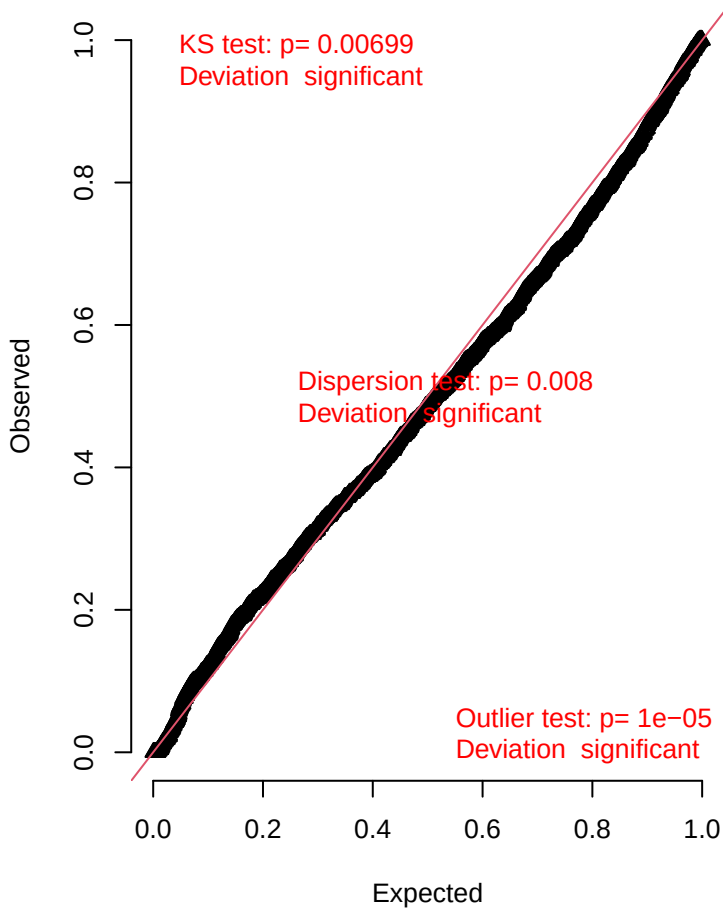
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

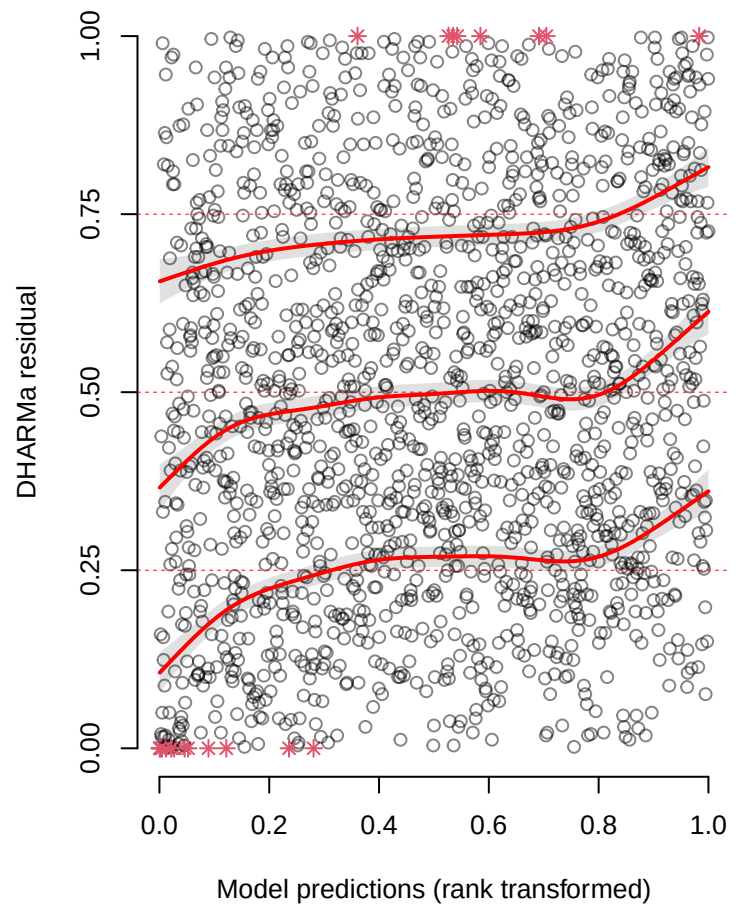


Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.031 | H1: Distance-based autocorrelation | p-value = 0.001

